COMMERCIAL ITEM DESCRIPTION

GROOVING MACHINE, SHEET METAL, HAND OPERATED, BENCH MOUNTED

The General Services Administration has authorized the use of this commercial item description in preference to type II of military specification MIL-F-23246A.

1. SCOPE

1.1 Scope. This description covers hand operated, bench mounted, sheet metal, grooving machines of the standard, heavy duty, commercial type, that are used in a production environment to close inside or outside longitudinal seams. The machines are intended for use in sheet metal shops and repair units.

1.2 Classification. The grooving machines shall be of the following sizes. The size to be furnished shall be as specified (see 6.1).

Size 1 - 22 gauge x 30 inch long capacity.
Size 2 - 22 gauge x 36 inch long capacity.
Size 3 - 20 gauge x 36 inch long capacity.
Size 4 - 20 gauge x 42 inch long capacity.

2. APPLICABLE DOCUMENTS

2.1 Documents. The following documents of the issue in effect on the date of invitation for bids or request for proposals form a part of this description to the extent specified herein.

Military Specification MIL-M-18058	- Machinery, Metal and Woodworking, Packaging of.
Federal Standard FED-STD-376	- Preferred Metric Units for General Use by the Federal Government.
American Society for Testing and Materials ASTM D 3951	- Standard Practice for Commercial Packaging.

3. SALIENT CHARACTERISTICS

3.1 Design. The grooving machine shall be new and of the manufacturer's commercial design in effect on the date of the solicitation. The grooving machine shall be capable of grooving or closing longitudinal seams in sheet metal

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cylinders of the thickness and lengths as specified in table I for the machine specified. The grooving machine shall meet the requirements specified herein and shall conform to current commercial industrial practice.

TABLE I. Grooving machine size and capacity.

Size	Gauge number	Maximum rated capacity in mild steel: Manufacturer's standard gauge in mild steel (inches)	Diameter of horn (maximum) (inches)	Working length (minimun) (inches)
1	22	0.0299	2	30
2	22	0.0299	2	36
3	20	0.0359	3	36
4	20	0.0359	3-1/2	42

NOTE: Unless otherwise specified value shall be not less than the value specified.

3.2 Frame and base. The frame and base shall be constructed of steel or cast iron of such size and weight to provide support for the overarm, horn, carriage, and operation mechanisms and provide the alignment necessary for grooving operations.

3.3 Horn. The horn shall be of finished steel with three longitudinal grooves for inside seams and a flat surface for outside seams. The horn shall be designed for rotating to desired position and have provisions for locking in the frame. A latch shall be provided to support the outer end of the horn. The latch shall be designed so it can be released for inserting the workpiece and shall be of sufficient strength for operating at rated capacity of the grooving machine. The horn shall be provided with stops for positioning of the workpiece.

3.4 Overarm. The overarm shall be made of cast iron or steel and shall support the carriage and latch arm.

3.5 Carriage. The carriage shall travel on the overarm and shall be driven by either an endless chain or rack and gear mechanism. The carriage shall hold the grooving and flattening rolls and shall be provided with adjusting devices to adjust the pressure to conform to metal thicknesses. Grooving machines of 20 gauge capacity and heavier shall be provided with a device to reverse the travel of the carriage.

3.6 Shafts and gears. The shafts shall be of machined steel and shall be fitted to suitable bearings. The gears shall be of suitable material and hardness and shall be either totally enclosed or provided with proper guards to protect the operator.

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3.7 Operating cranks. The grooving machine shall be provided with all necessary operating cranks and levers necessary for operation within its rated capacity.

3.8 Rolls. Unless otherwise specified (see 6.1), each grooving machine shall be furnished with two each flattening rolls and two each grooving rolls.

3.9 Measurement system. All dimensions in this description are given in US units. These dimensions may be converted to SI units through the use of conversion factors and methods specified in FED-STD-376.

3.10 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials to the maximum extent practical.

3.11 Workmanship. Workmanship of the grooving machines shall be of a quality equal to that of comparable products furnished to the commercial market.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection, examination, and test requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections, examinations, or tests set forth in this description where such inspections, examinations, and tests are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Examination. Each grooving machine shall be visually examined to determine conformance with all requirements of this description.

4.3 Operational test. Each grooving machine shall be operated without performing grooving operations to ensure operation of all moving parts and adjusting mechanisms.

4.4 Packaging inspection. Packaging shall be inspected for conformance with section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified in the contract or purchase order, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.1), level A preservation, level A or level B packing, and marking shall be in accordance with MIL-M-18058.

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6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

a. Title, number, and date of this commercial item description.
b. Size, of grooving machine required (see 1.2).
c. Flattening rolls and grooving rolls, if different (see 3.8).
d. Packaging in accordance with MIL-M-18058,if required specify level of preservation and packing (see 5.1).

6.2 Cross reference data. Items covered by Military Specification MIL-F-23246A are cross referenced to items covered by this description as follows:

MIL-F-23246A	A-A-51169
Type II	(Deleted)
	Size 1 (New size)
	Size 2 (New size)
	Size 3 (New size)
	Size 4 (New size)
	Size 5 (New size)

MILITARY INTERESTS:

Custodians

Army - AL
Air Force - 99

Review Activities

Navy - YD
Air Force - 84
DLA - GS

CIVIL AGENCY COORDINATION ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

DLA - IP

Project 3441-0128

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-51169	Grooving Machine, Sheet Metal, Hand Operated, Bench Mounted

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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